

Introduction to neuronal modelling

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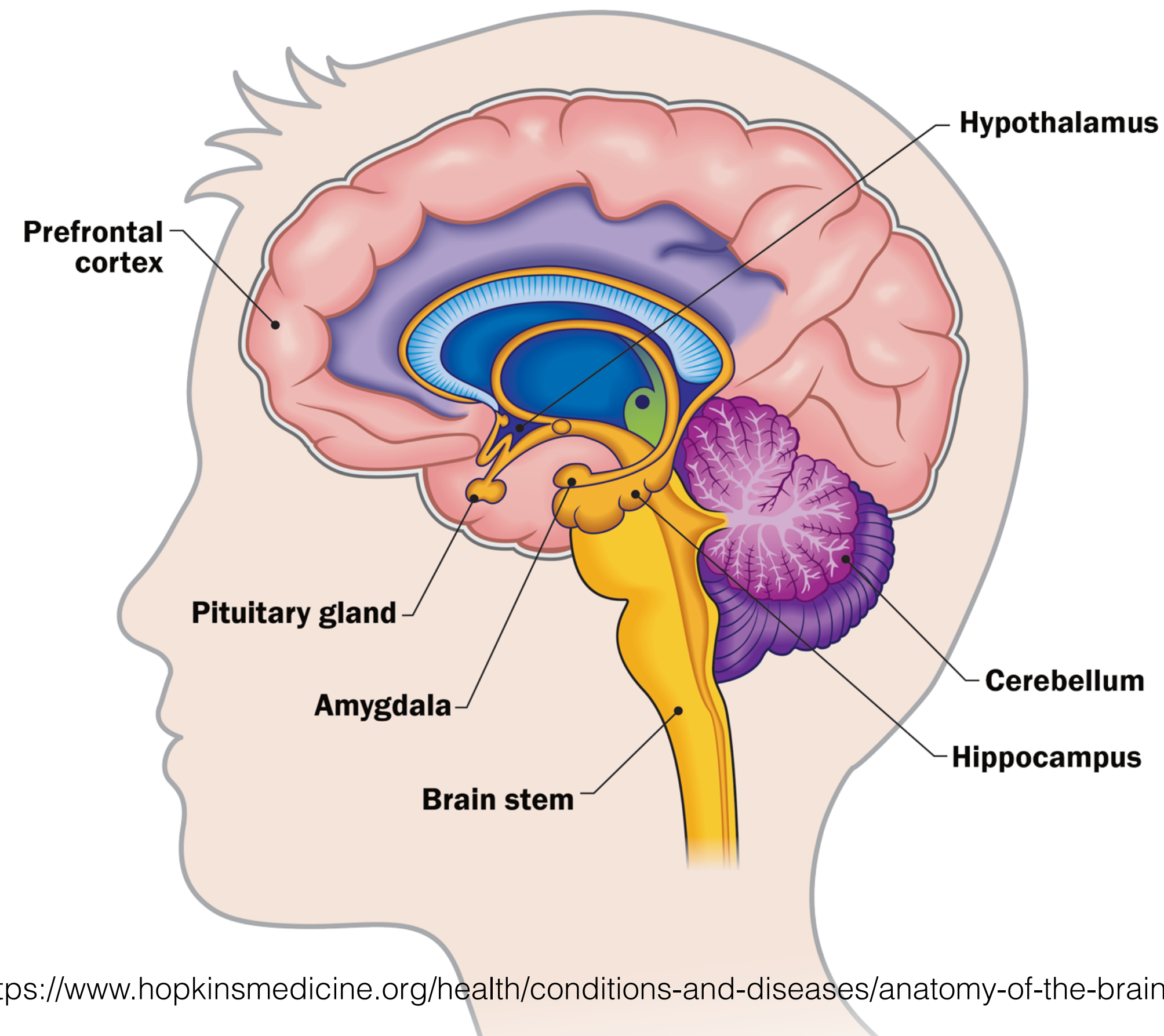
What we will cover

- How the brain works, in 5 minutes
- Building blocks of neuronal modelling
 - Single neuron models, Hodgkin-Huxley
 - Synapse models

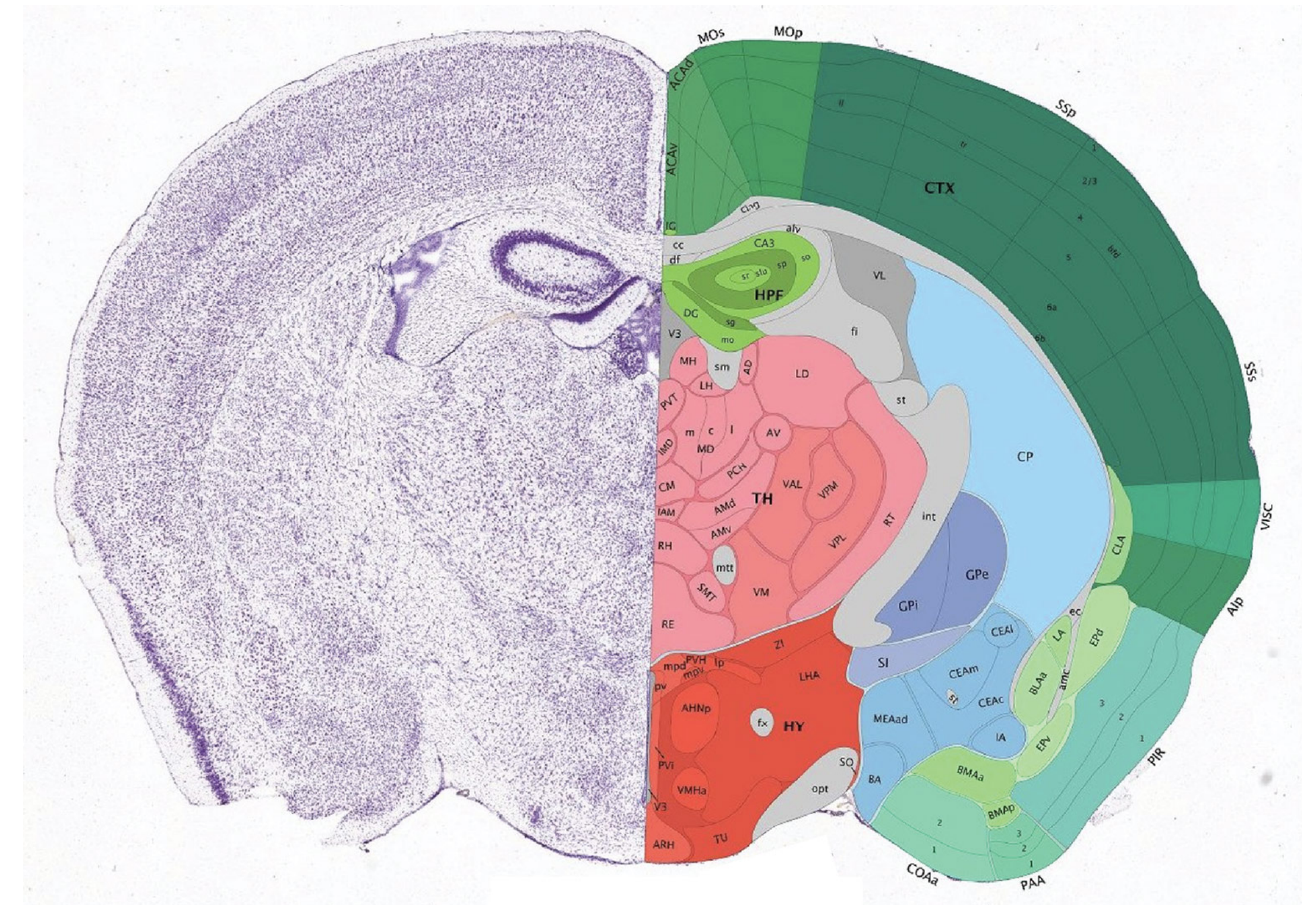
how the brain works

Brain regions

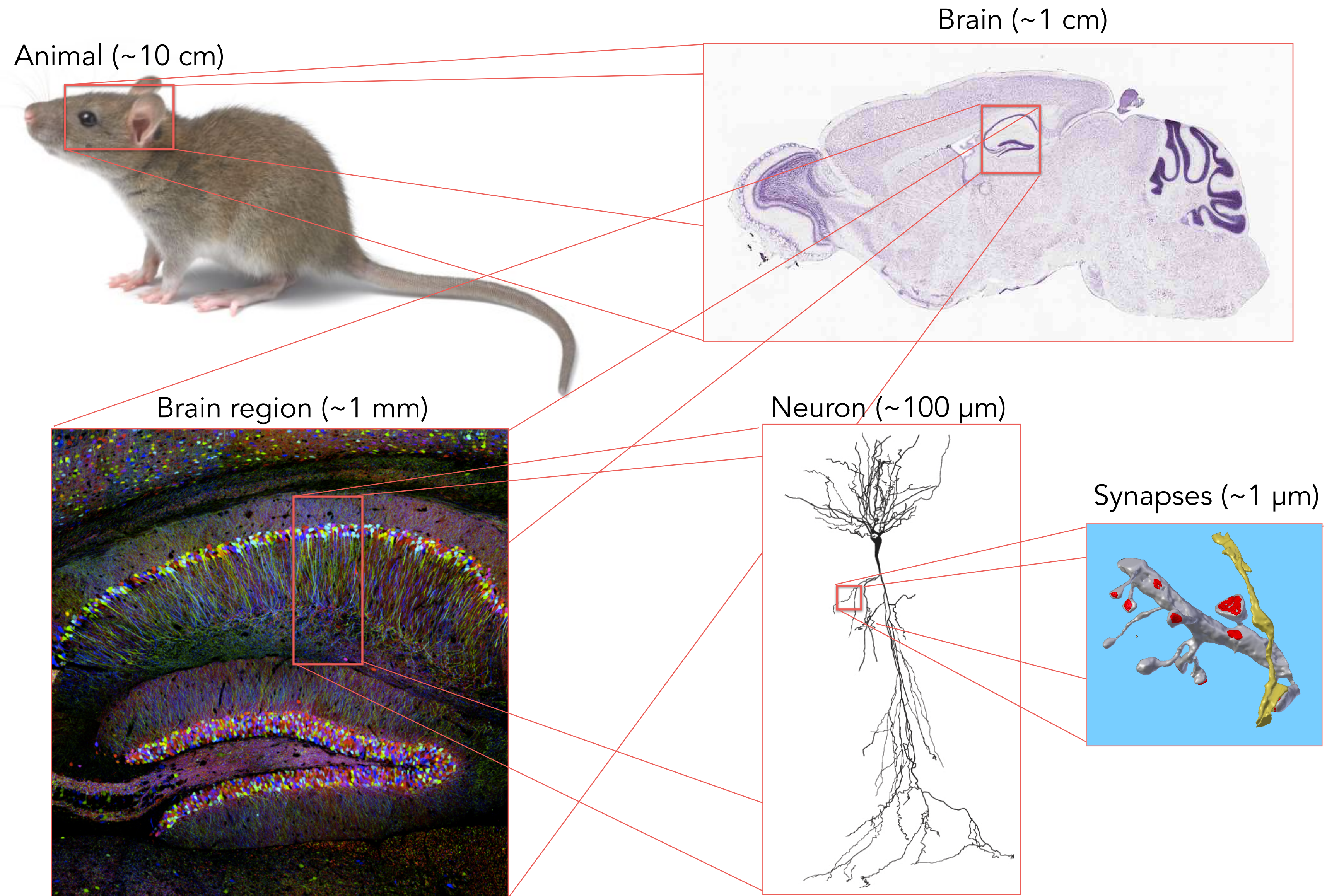
The brain is **modular**: it has tens of specialised regions that perform different tasks: sensory processing, motor control, learning & memory, decision making, emotional regulation, etc



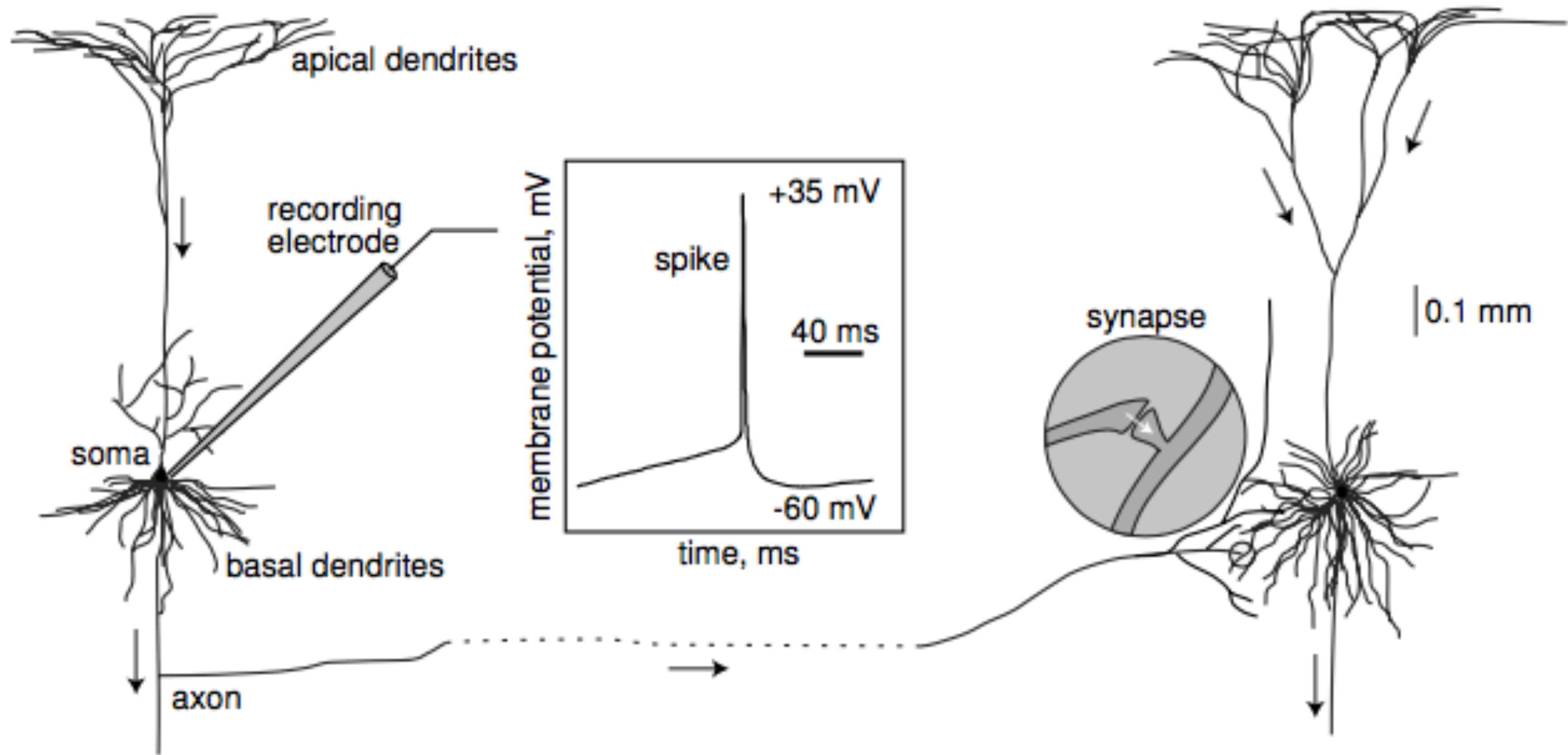
Mouse brain



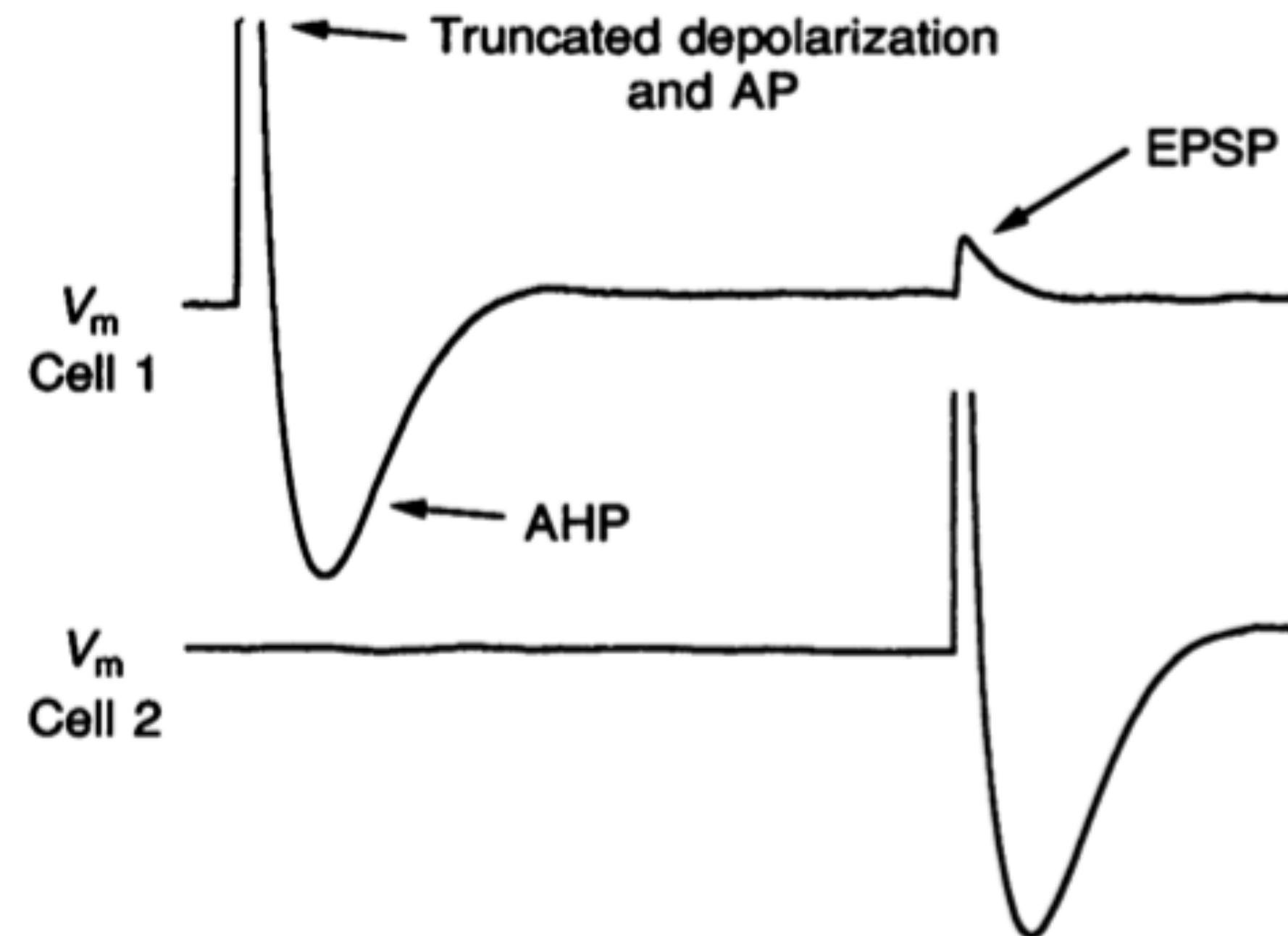
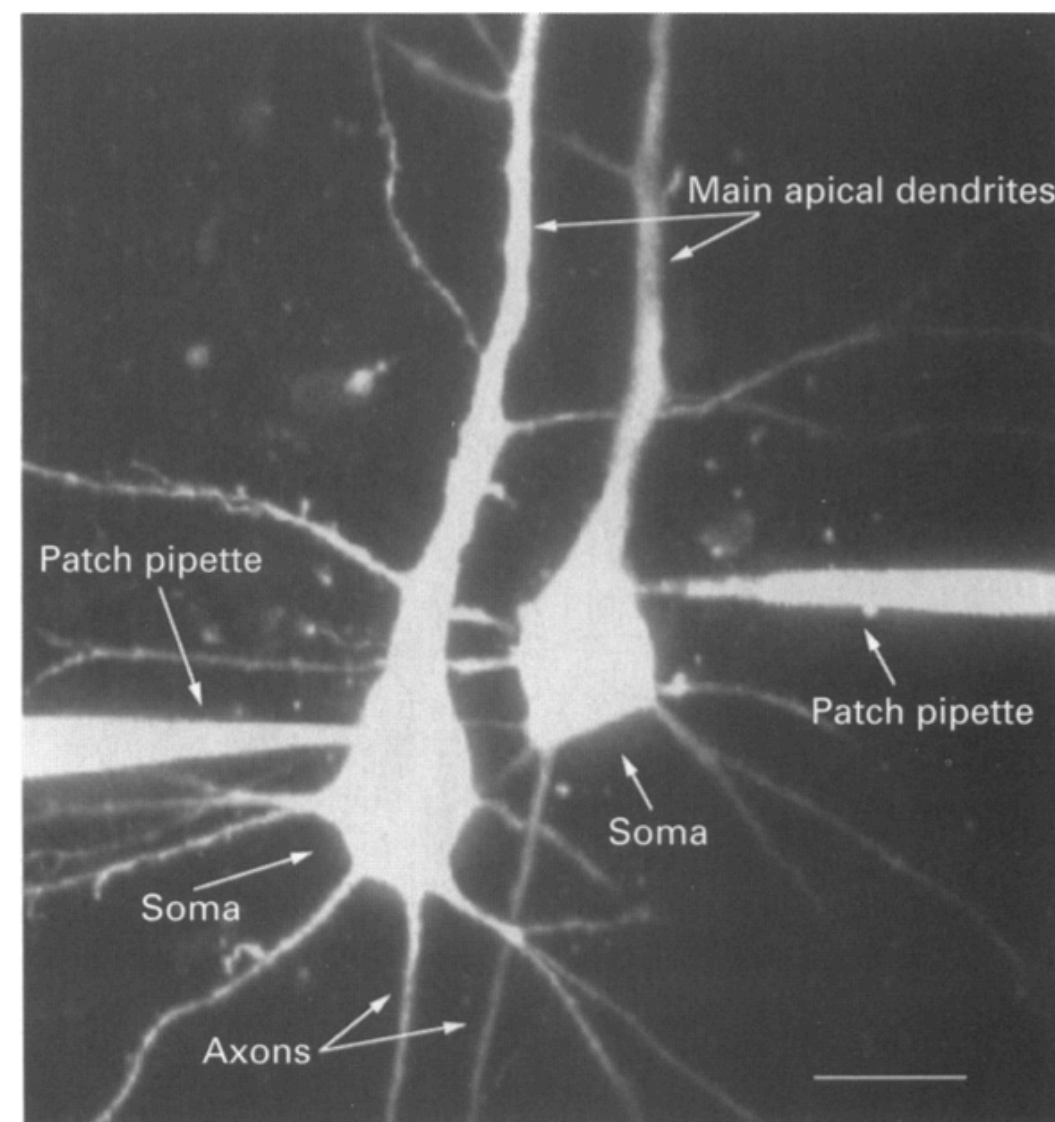
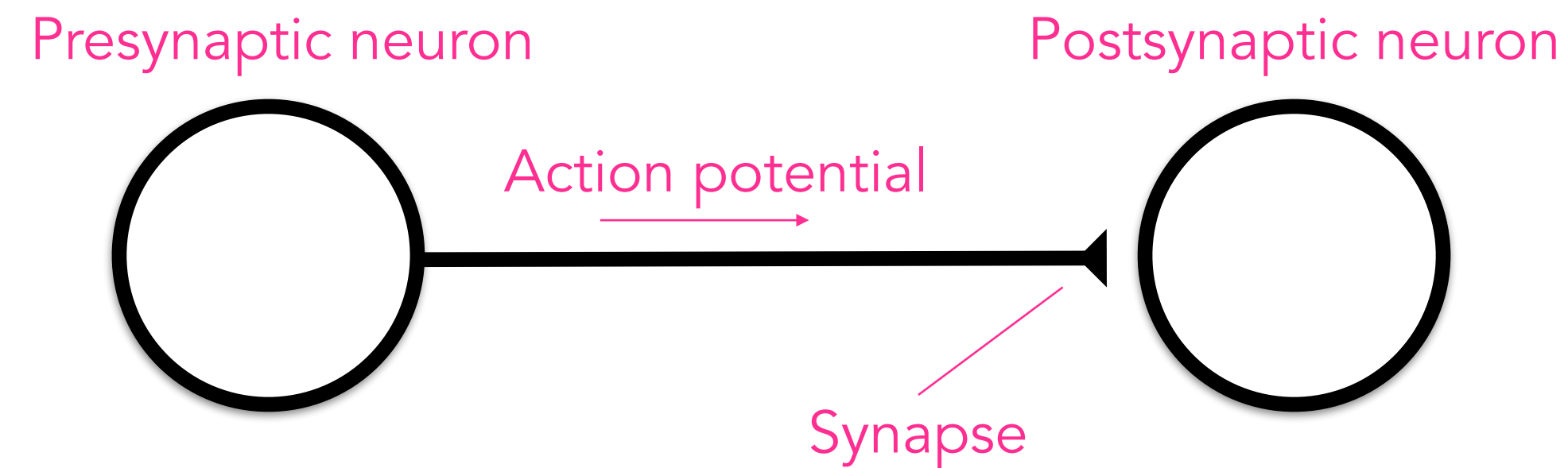
Spatial scales of the brain



How do brains work?



What is a synapse?



How do synapses work?

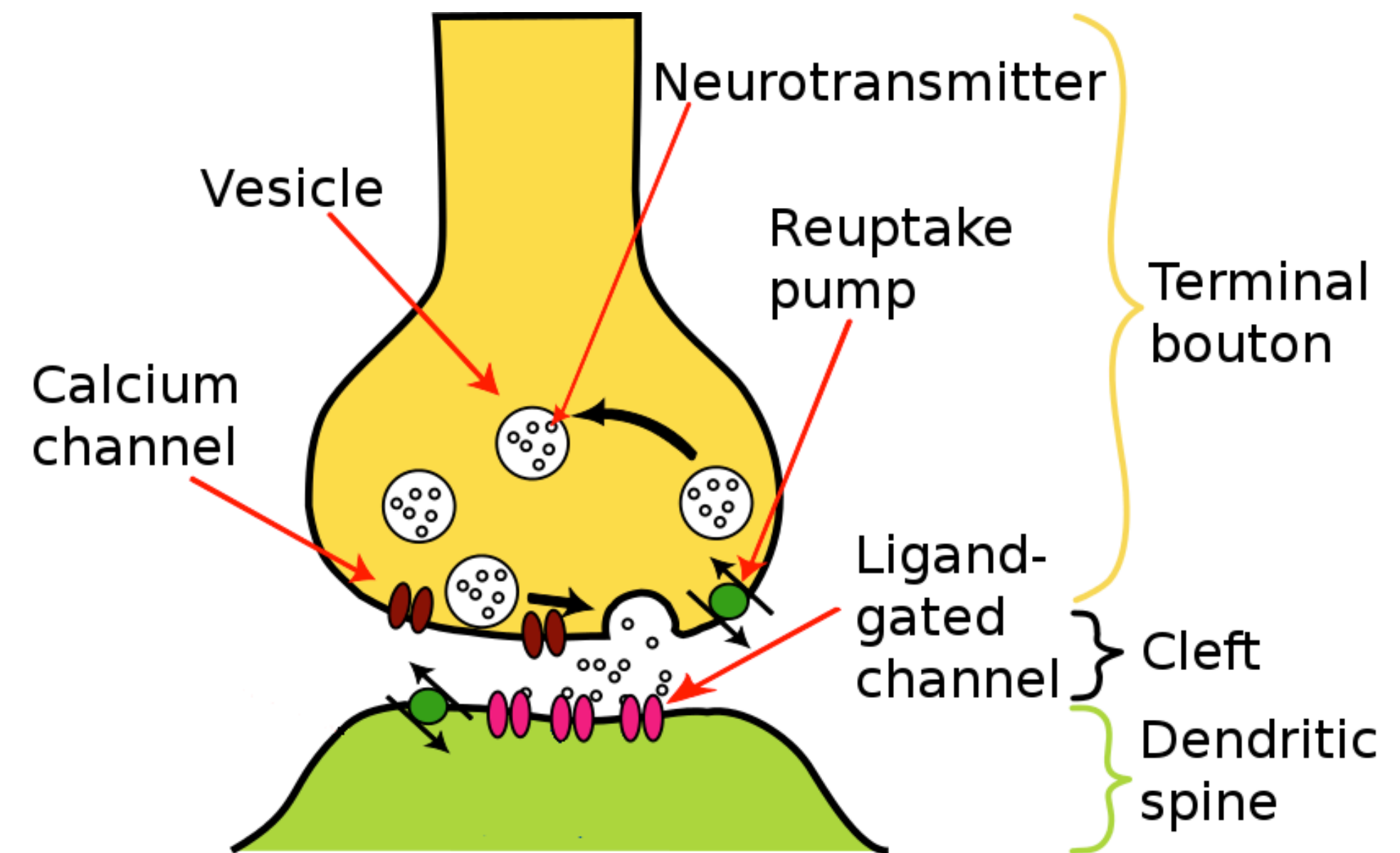
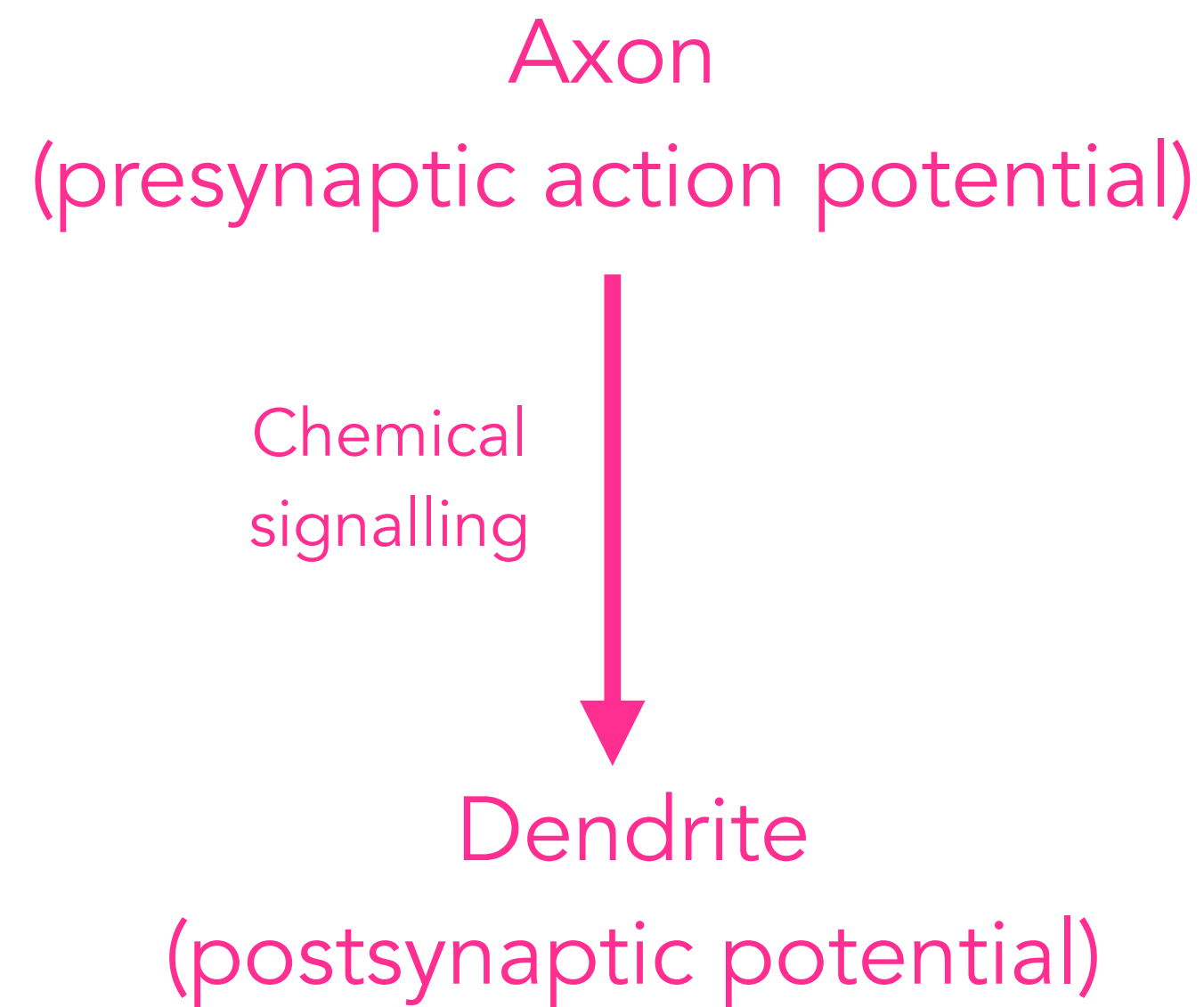


Image from Wikipedia (modified by C Houghton)

Summary on brain basics

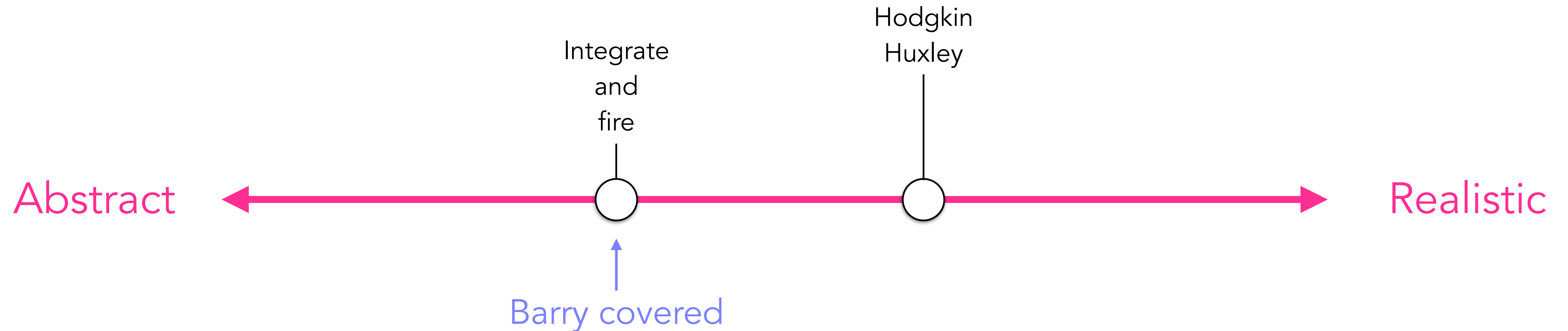
- The brain is a large network of neurons (and glia), with many specialised **regions** at a macroscale, and lots of **cell types** and **detailed wiring patterns** at a microscale.
- It is an **electro-chemical computer**.
- Each **neuron** uses its membrane **voltage to integrate incoming signals**, then sometimes broadcasts a **spike** output signal (when the voltage crosses a threshold).
- **Synapses** are the **connections** between neurons (usually unidirectional).

Building blocks of brain models

Building blocks of brain models

Neurons

Models of single neurons



The Hodgkin-Huxley model

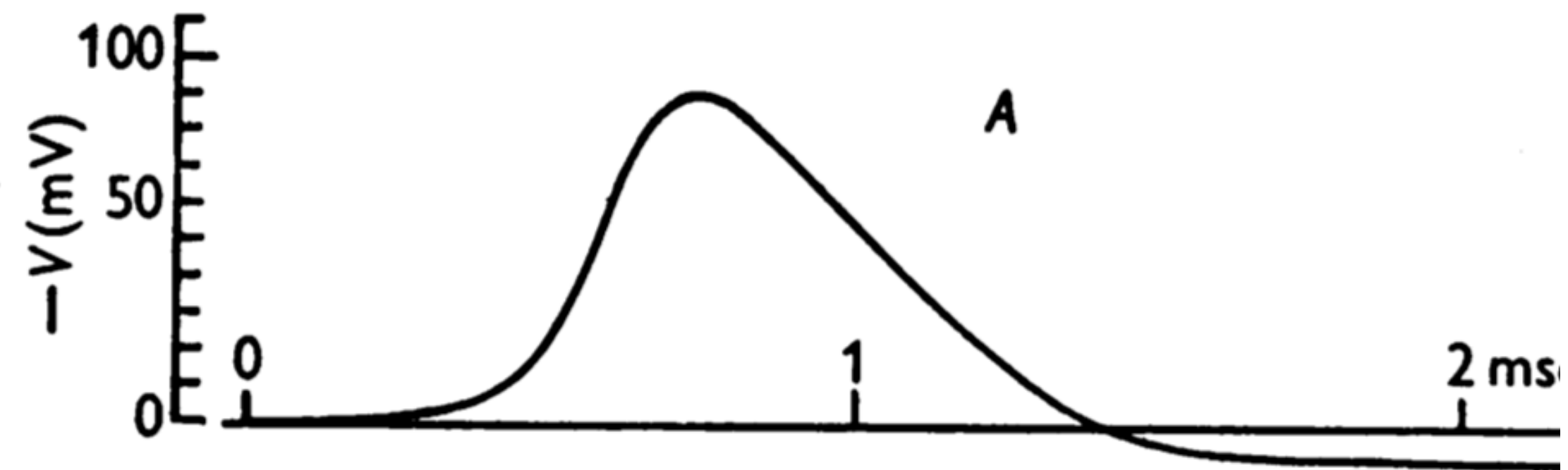
- The original Hodgkin-Huxley model is a mathematical model of the electrical dynamics of the 'giant' axon of the squid *Loligo forbesi*.
- Its key success was to demonstrate that **two voltage-gated membrane conductances** were sufficient to explain the **action potential**.
- These days people often use the term "Hodgkin-Huxley style model" more loosely to mean any mathematical model of any neuron that is built using **conductance-based** dynamics.
- The Hodgkin-Huxley model stands as one of the outstanding successes of computational neuroscience.



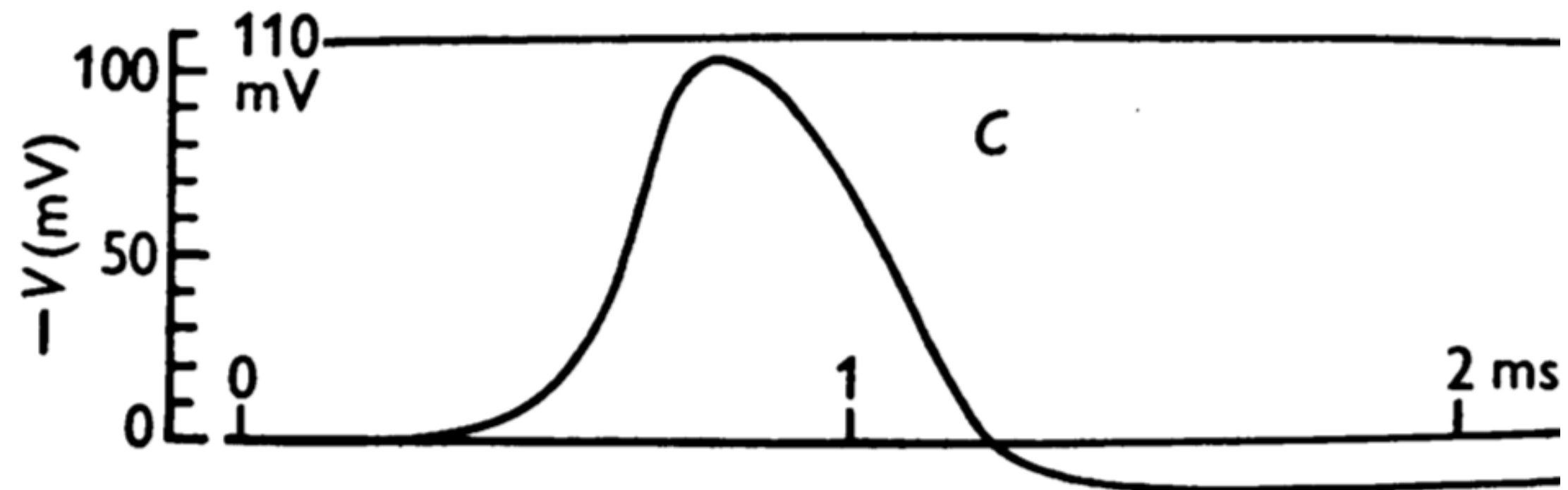
Loligo forbessi



Model

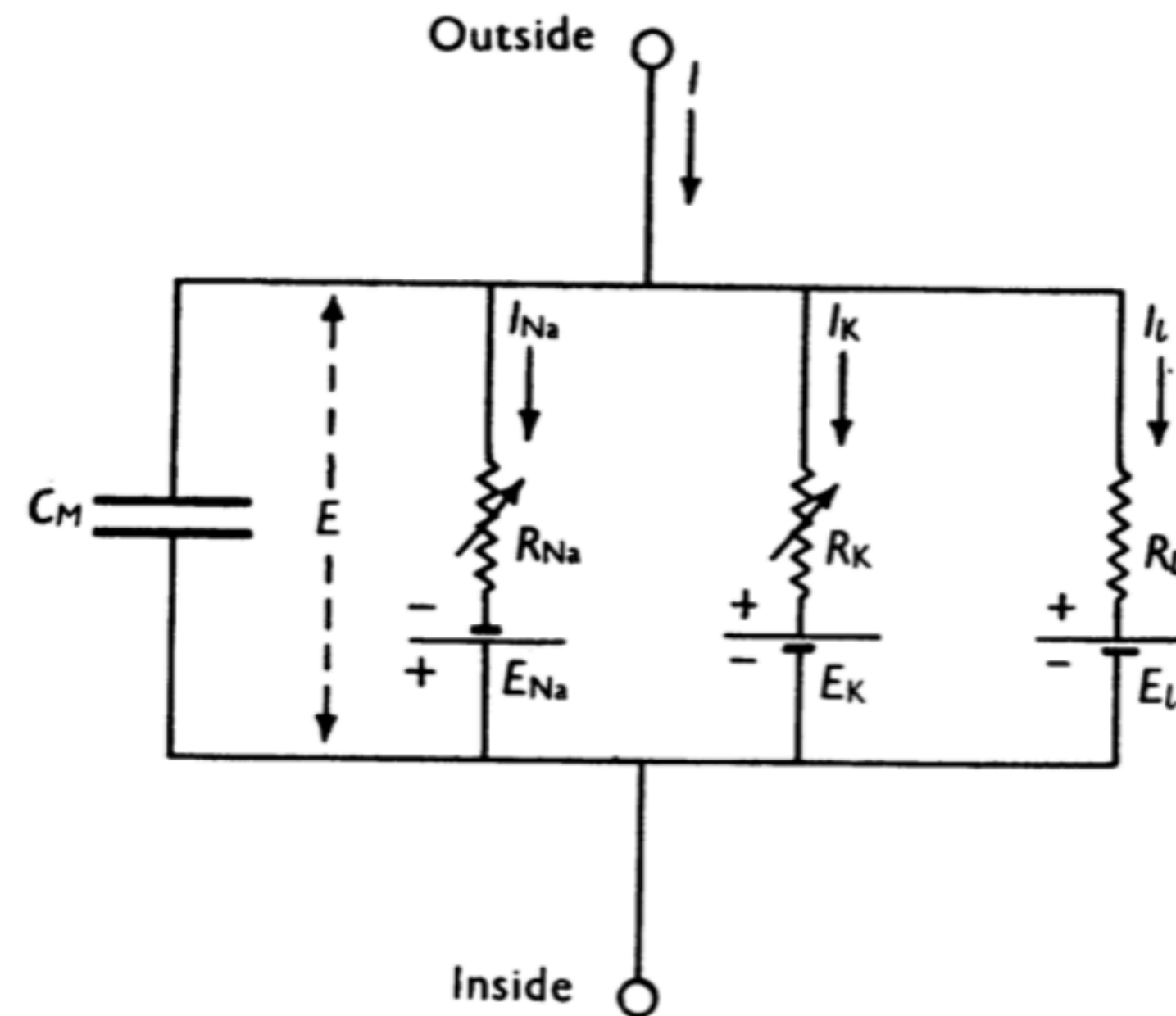


Data



What does the model consist of?

The HH model



$$C_M \frac{dV}{dt} = I_{Na} + I_K + I_l$$

$$I_x = g_x (E_x - V) \quad \dots \text{where } x \text{ is } Na, K \text{ or } l$$

$$g_x = ?$$

How do we model the conductances?

How do we model the conductances?

Using time and voltage-dependent gating variables.

$$g_{Na} = \bar{g}_{Na} m^3(V, t) h(V, t)$$

$$g_K = \bar{g}_K n^4(V, t)$$

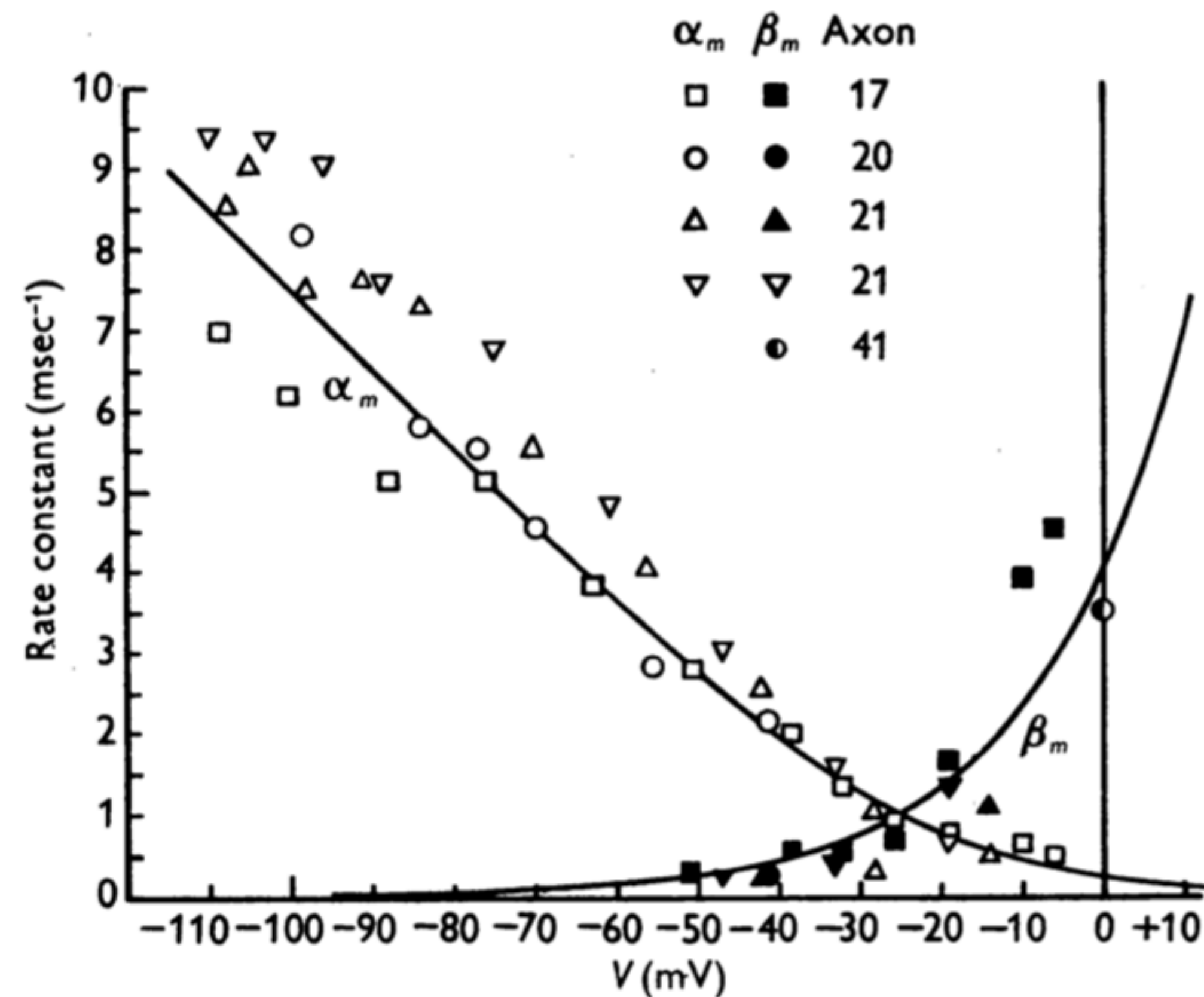
How do the gating variables evolve in time?

$$\frac{dm}{dt} = \frac{m_{\infty}(V) - m}{\tau_m(V)}$$

How do the steady-state values and time constants depend on voltage?

$$m_{\infty}(V) = \frac{\alpha_m(V)}{\alpha_m(V) + \beta_m(V)} \qquad \tau_m(V) = \frac{1}{\alpha_m(V) + \beta_m(V)}$$

How do the forward and backward rate constants depend on voltage?
Hodgkin and Huxley fit them to match their voltage-clamp data.



$$\alpha_m(V) = \frac{0.1(V + 40)}{1 - e^{-(V+40)/10}}$$

$$\alpha_h(V) = 0.07e^{-(V+65)/20}$$

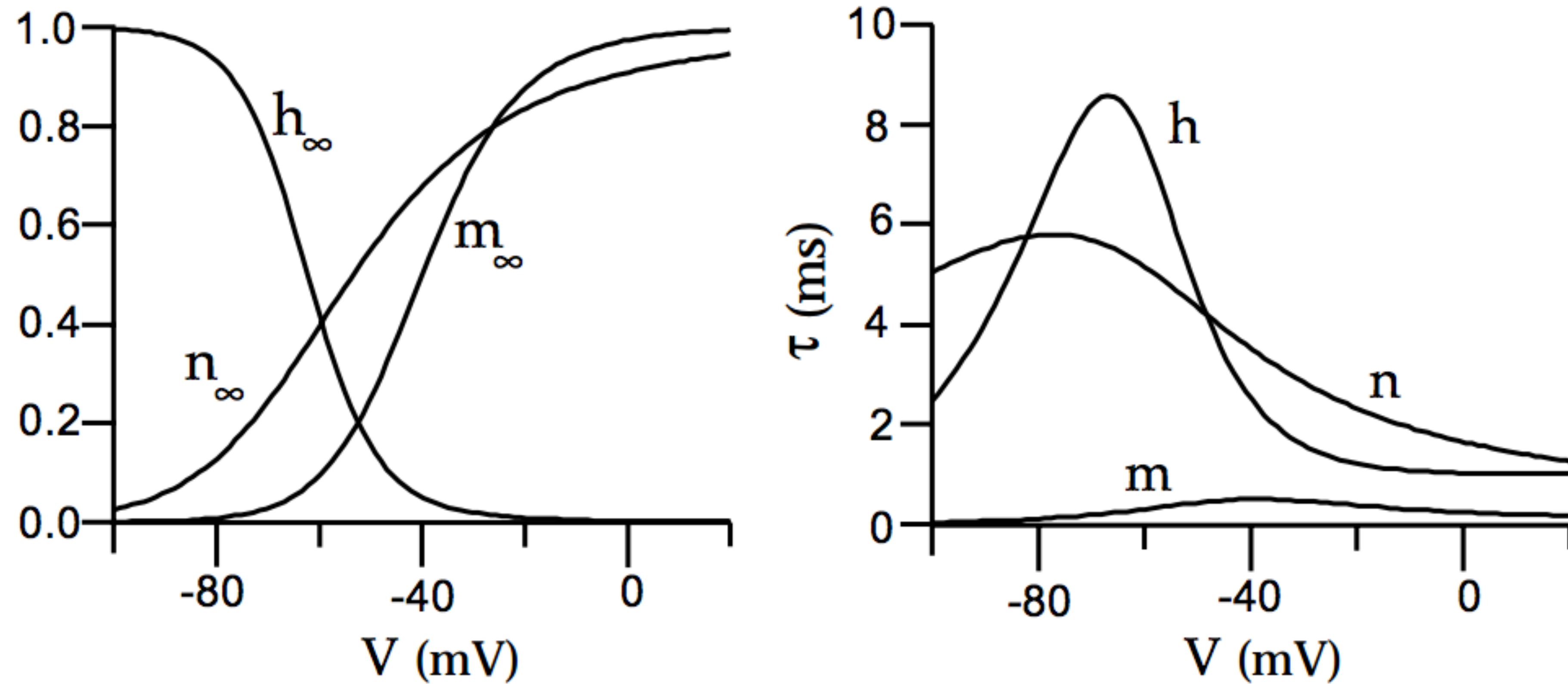
$$\alpha_n(V) = \frac{0.01(V + 55)}{1 - e^{-(V+55)/10}}$$

$$\beta_m(V) = 4e^{-(V+65)/18}$$

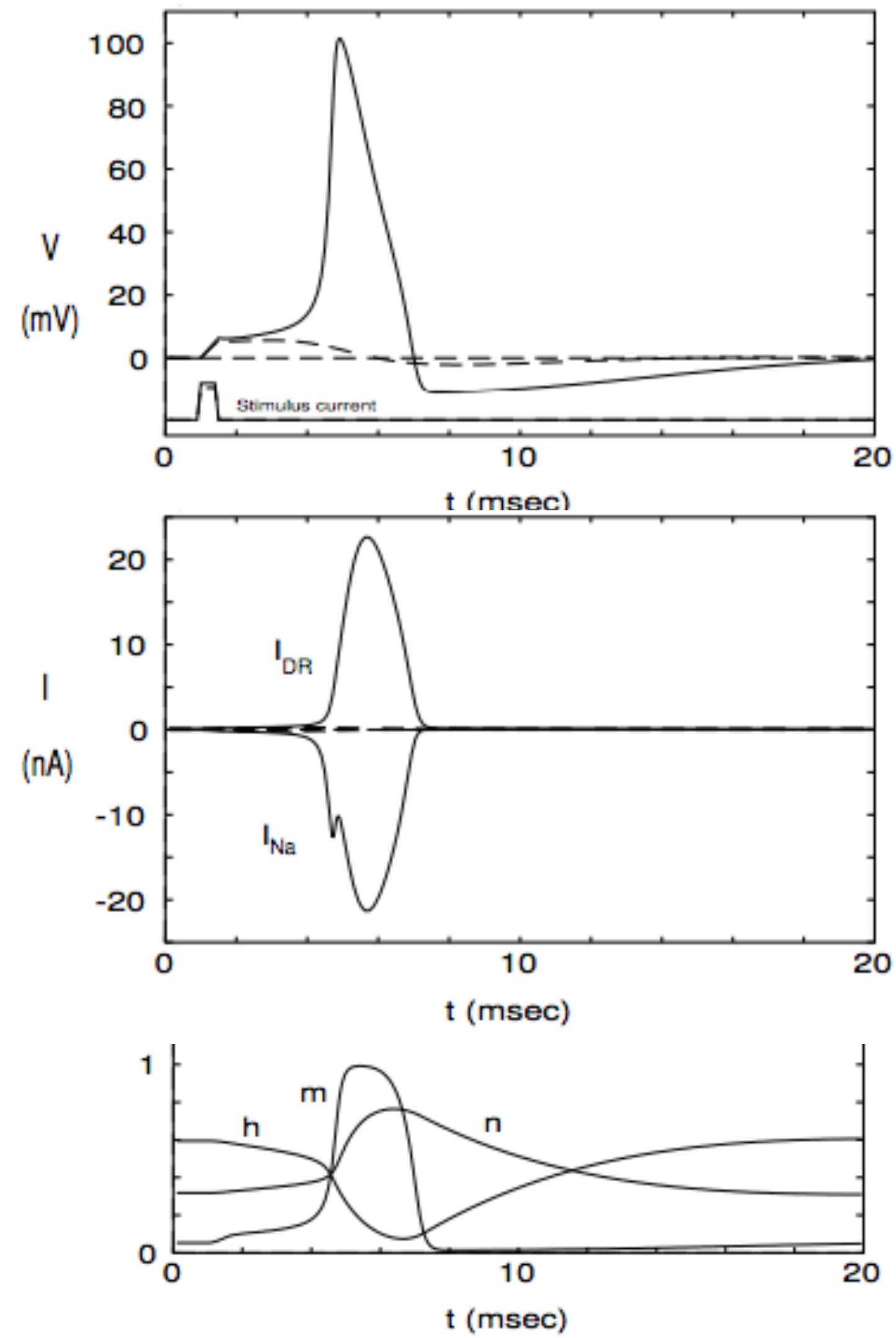
$$\beta_h(V) = \frac{1}{1 + e^{-(V+35)/10}}$$

$$\beta_n(V) = 0.125e^{-(V+65)/80}$$

Gating variables steady-state values and time constants as a function of voltage



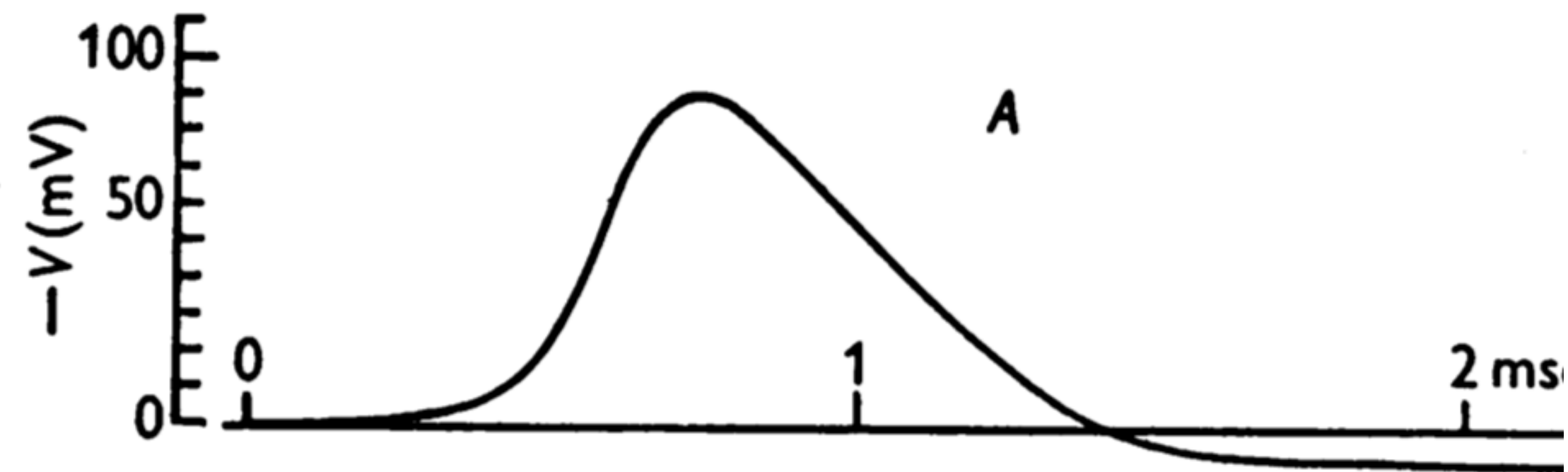
What does the HH model do?



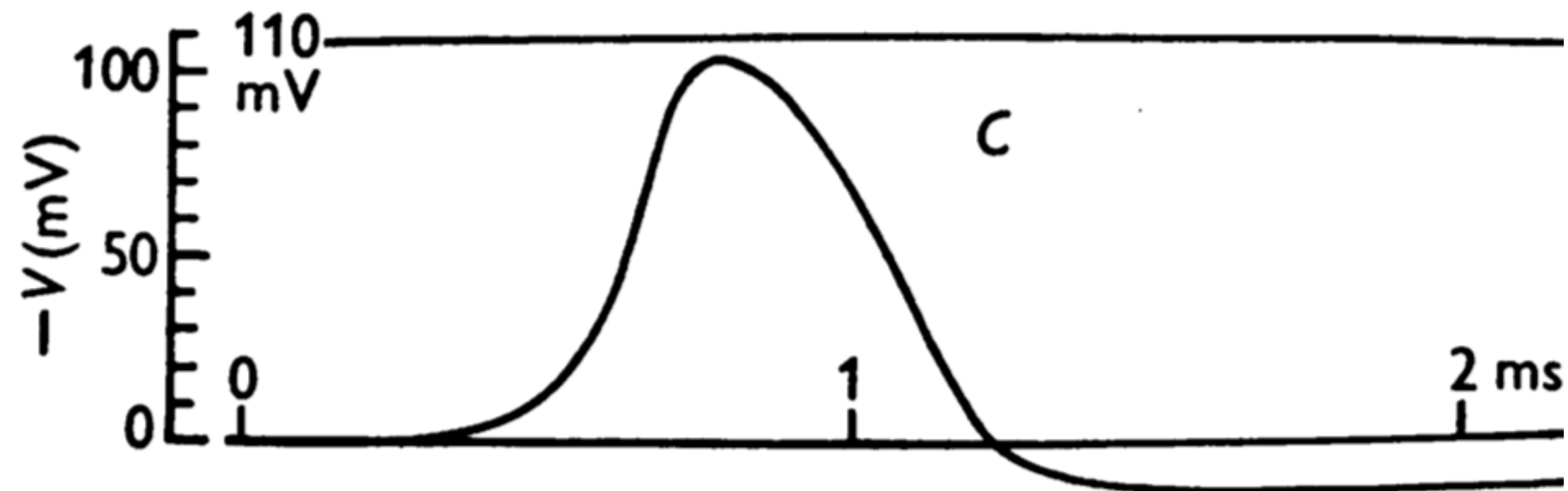
Koch (1999)

What does the HH model do?

Model



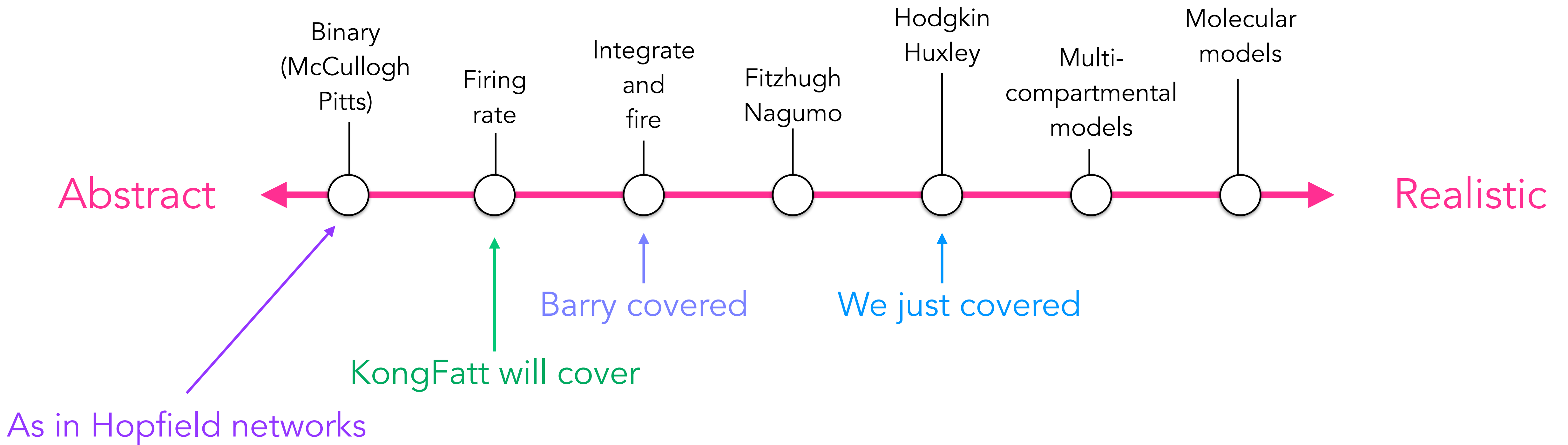
Data



What does the HH model *not* do?

- It is unlike the action potentials in mammalian neurons:
 - different ion channels
 - different waveform
 - energy inefficient
- Not a good model for myelinated axons
- It is deterministic and continuous. We now know that real ion channels are discrete and stochastic.
- Assumption of multiple independent gates per channel type is biophysically unrealistic.

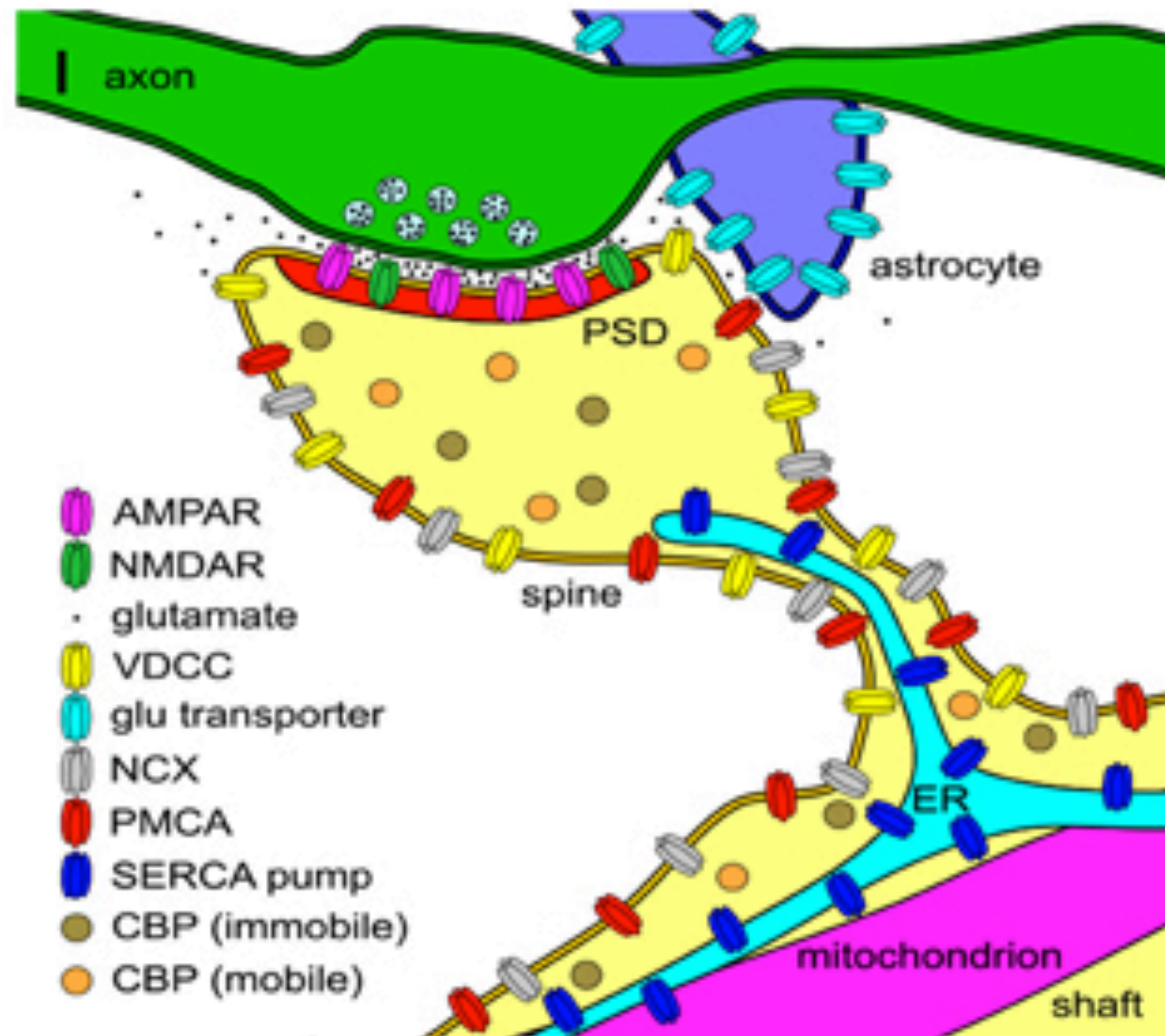
Models of single neurons



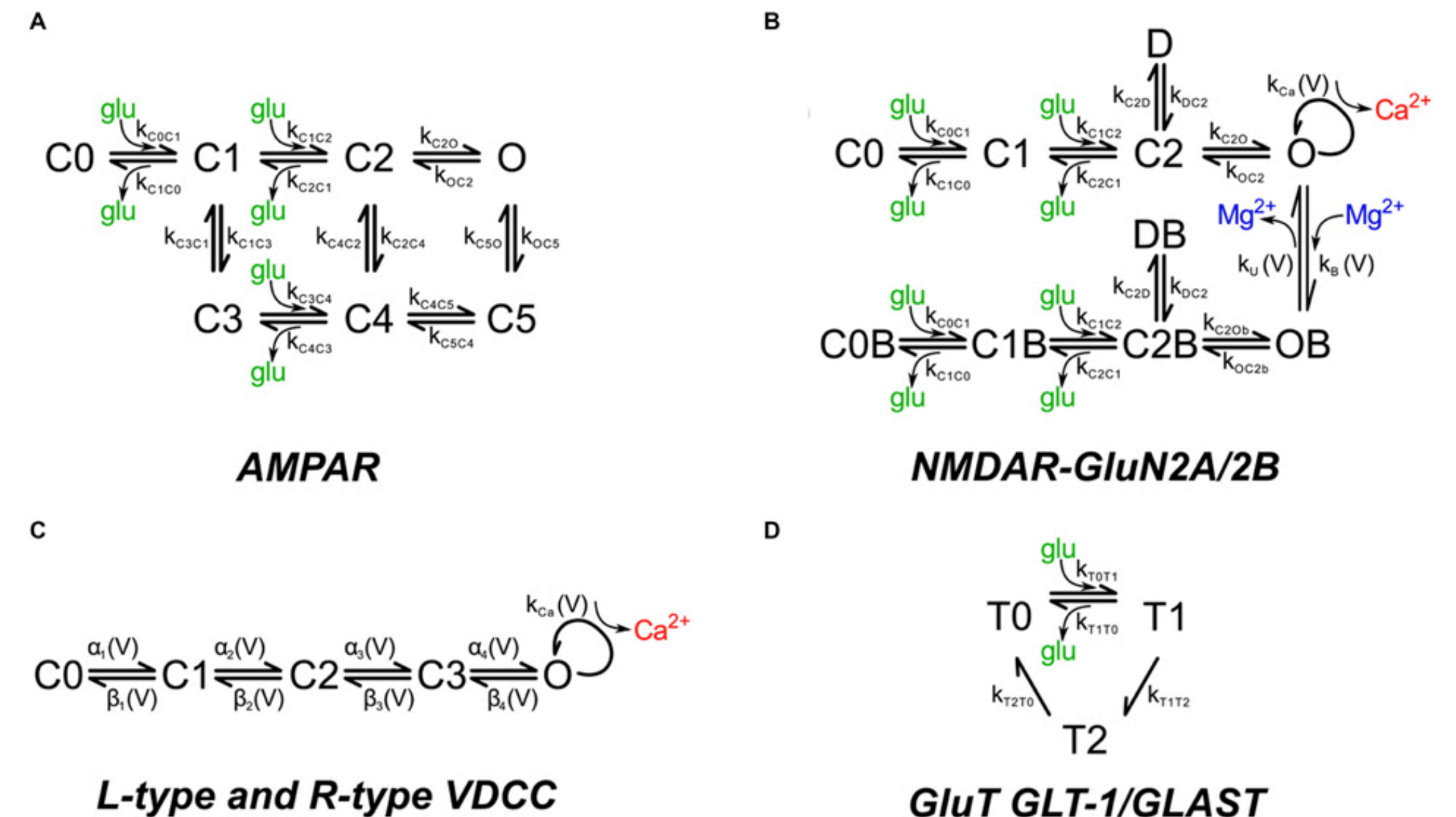
Building blocks of brain models

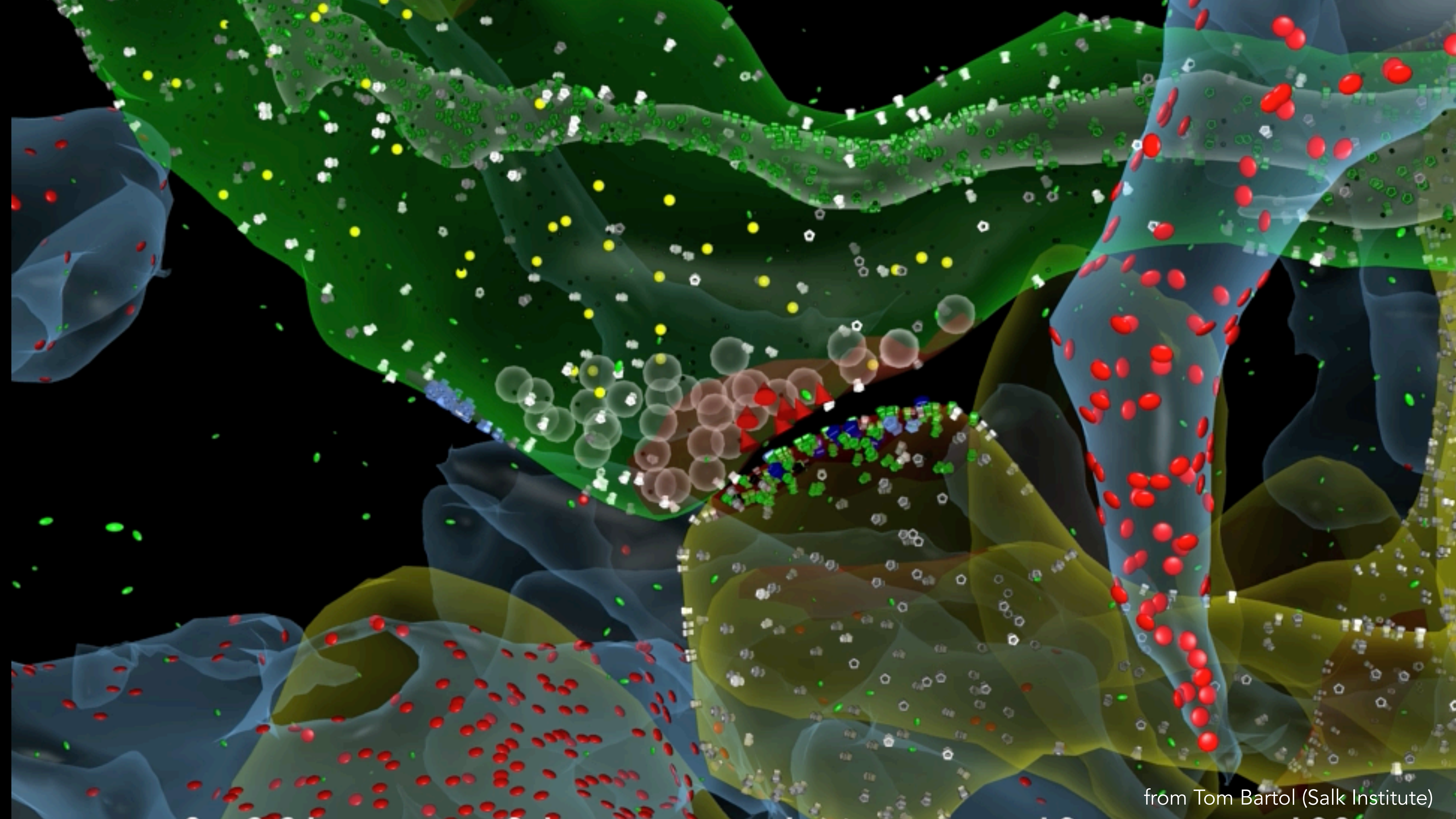
Synapses

MCell simulation of synaptic release



Bartol et al, *Frontiers Syn Neuro* (2015)





from Tom Bartol (Salk Institute)

How can we computationally model a synapse?

- We *could* simulate the dynamics of each molecule involved in the signalling process (like the MCell simulation).
- But since that is very computationally *expensive*, we might instead go for a reduced mass-action chemical-kinetics model.
- However a lot of people still find even that too expensive and parameter-heavy, so instead use even simpler phenomenological models that black-box the synapse as a simple input-output system.

Simple synapse models

The most common way to phenomenologically model a synapse is as a time-dependent conductor in series with a battery.

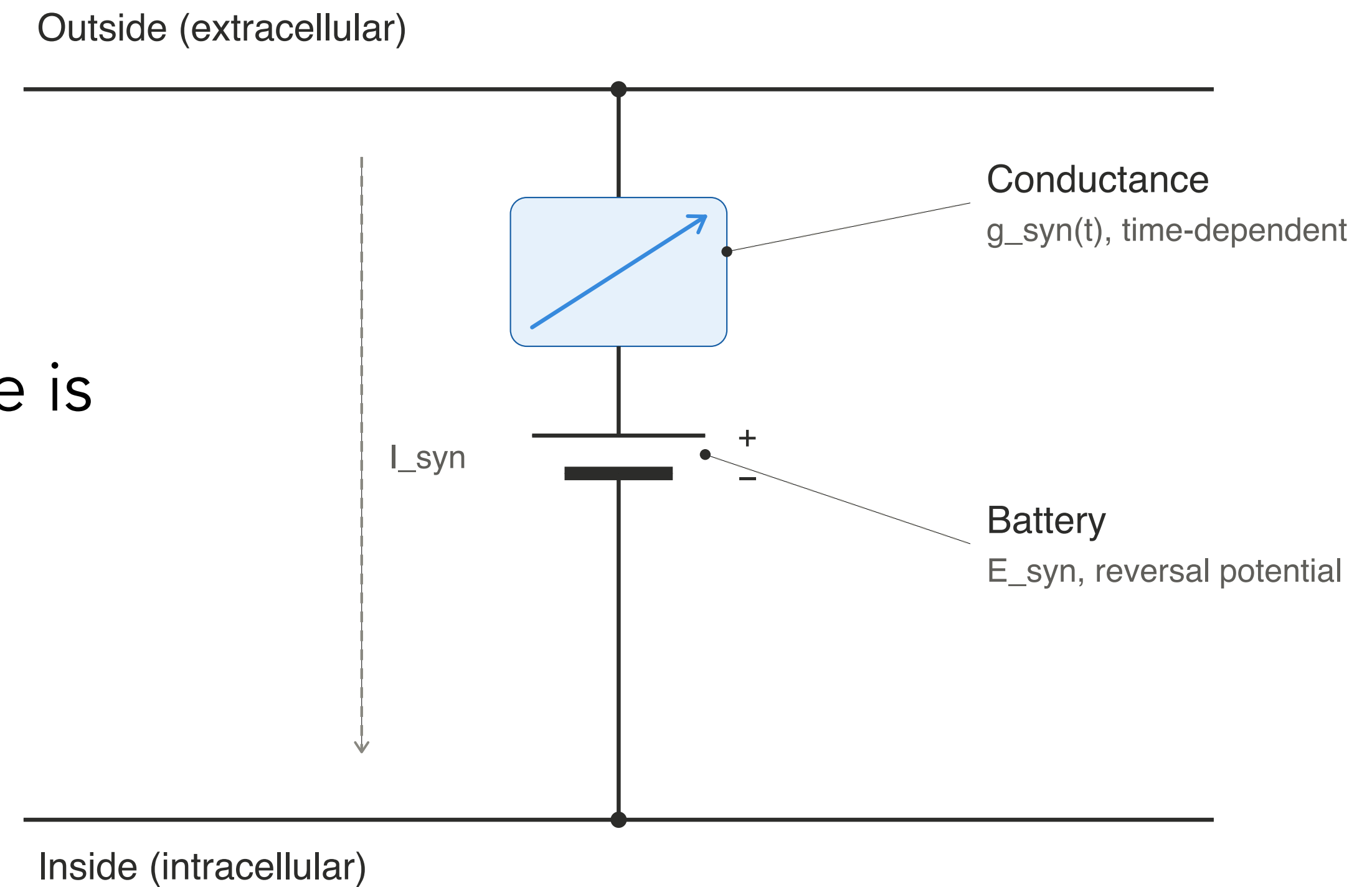
$$I_s(t) = \bar{g}_s s(t) (E_s - V)$$

The value of E_s determines whether the synapse is excitatory or inhibitory:

for excitatory synapses E_s usually = 0 mV

for inhibitory synapses E_s usually = V_{rest}

But how should we model $s(t)$?

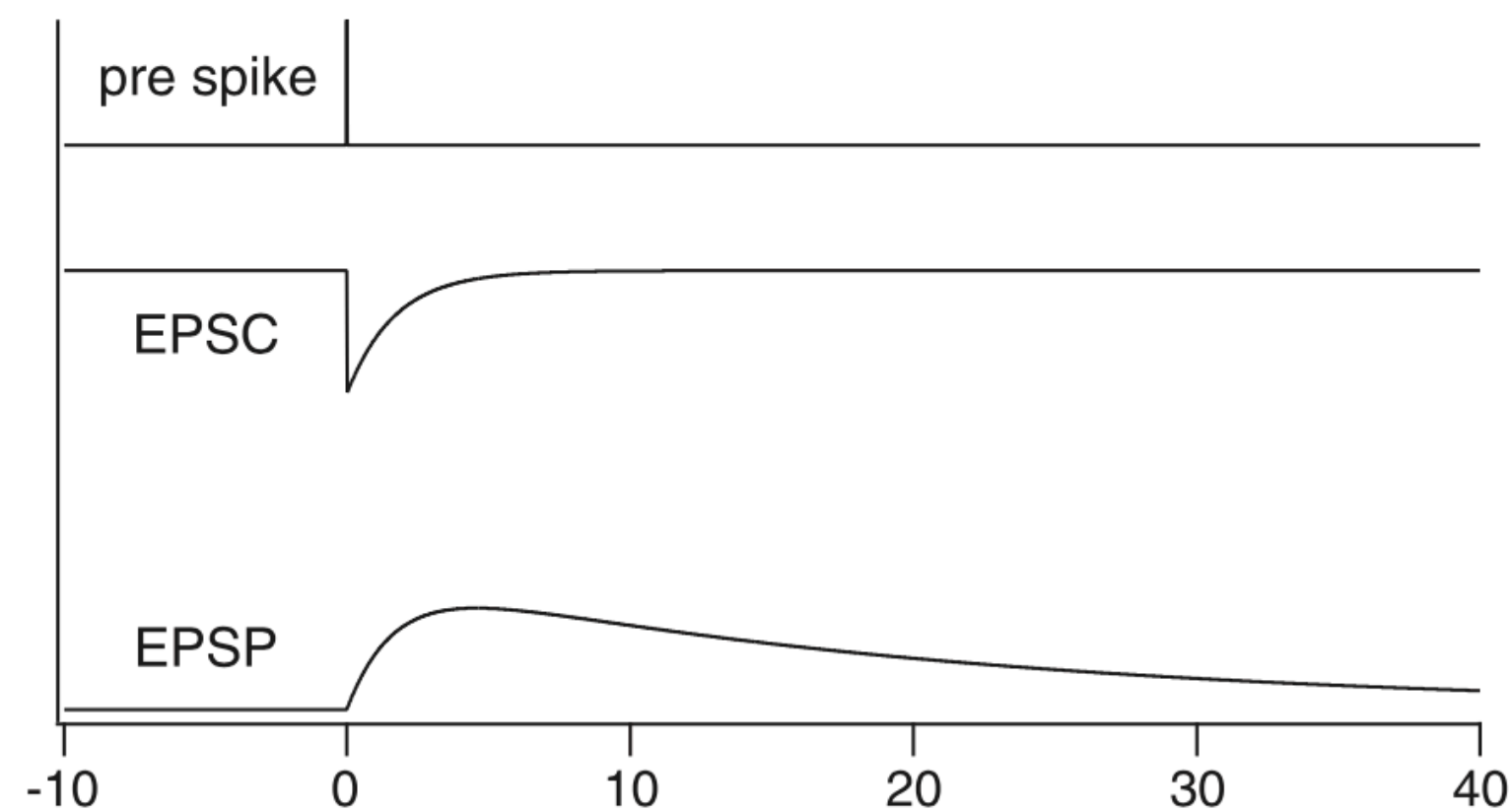


Simple synapse models

Single exponential

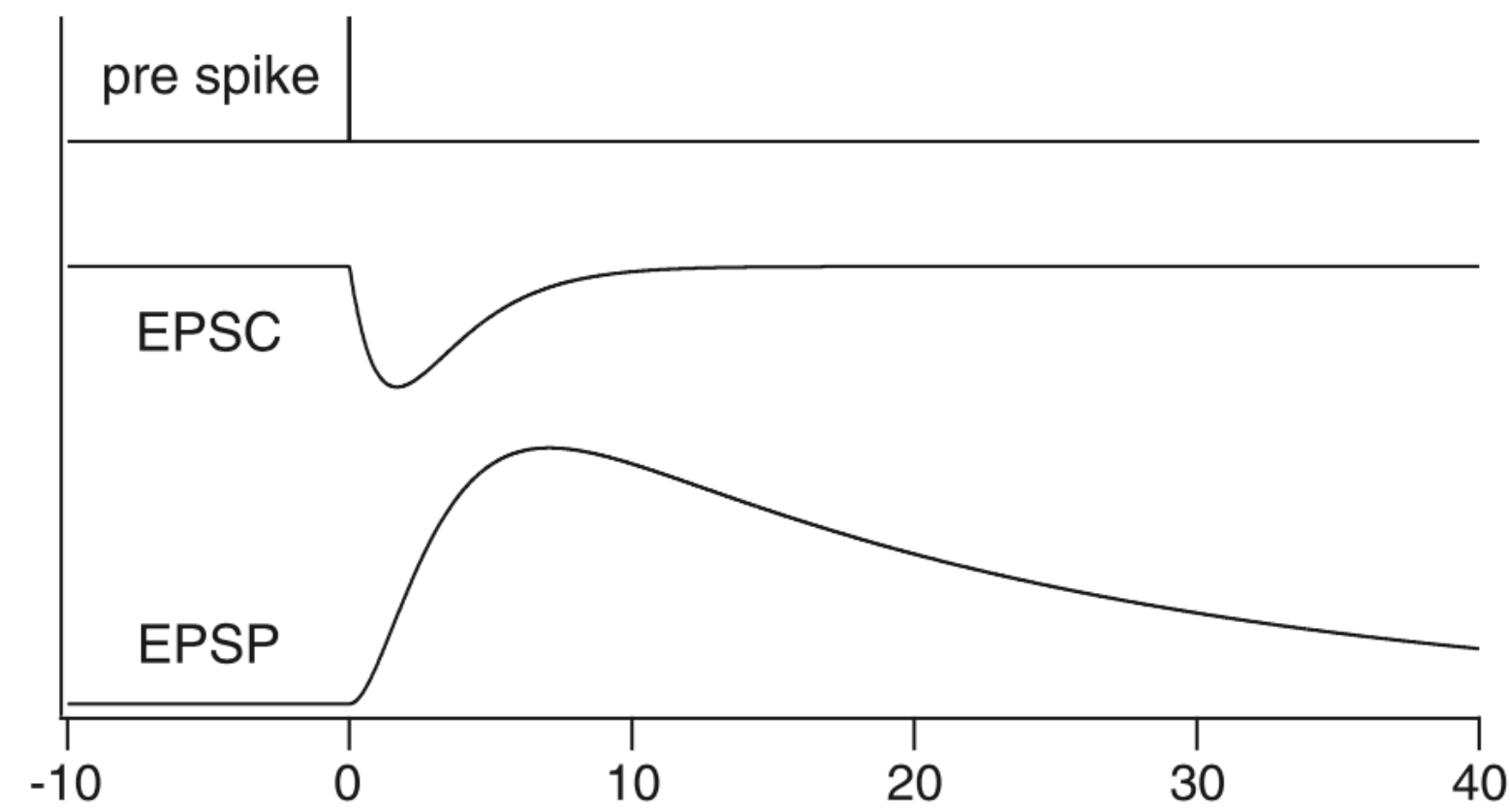
$$s(t) \rightarrow s(t) + 1$$

$$s(t) = e^{-t/\tau_s}$$



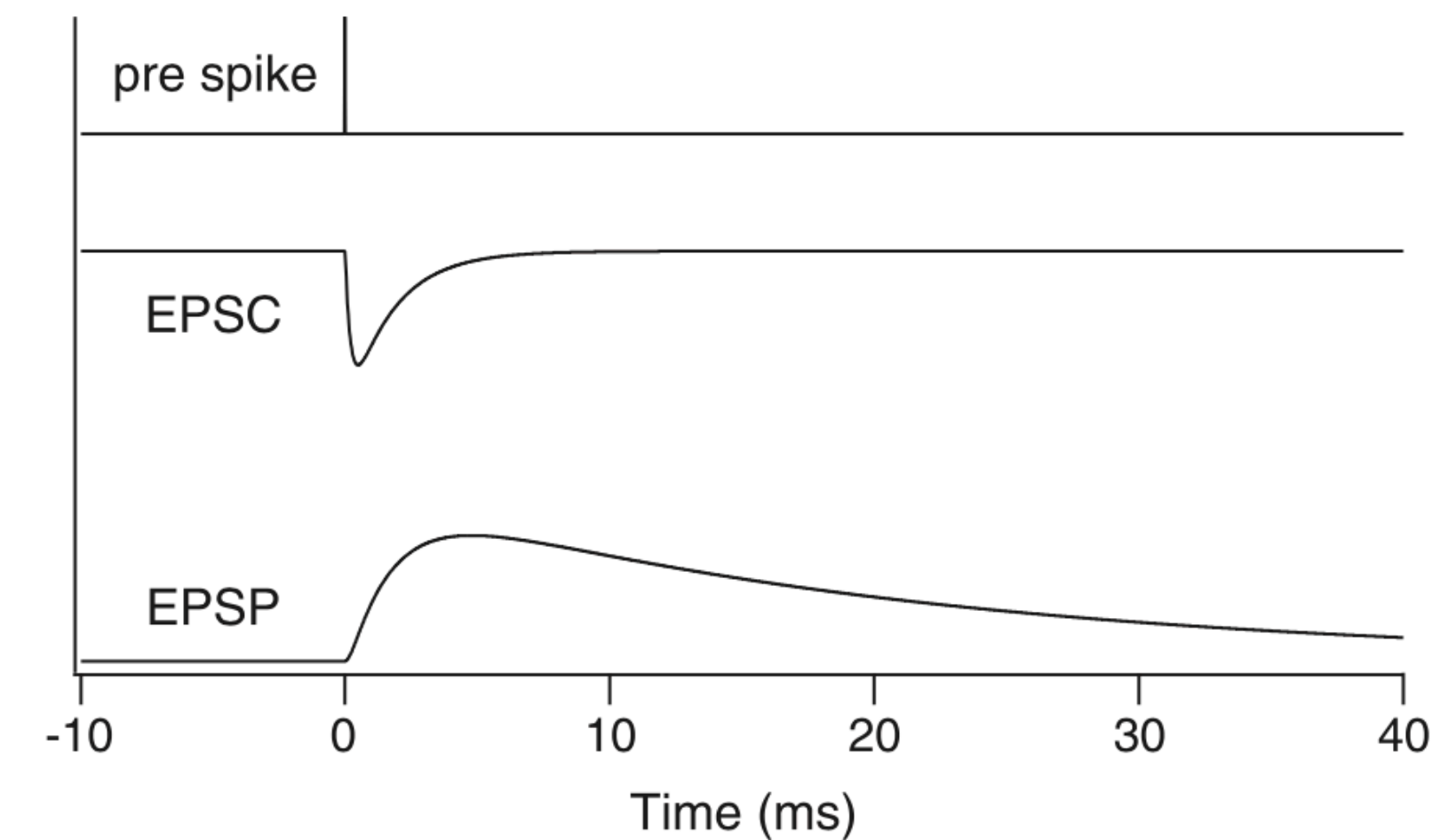
Alpha function

$$s(t) = te^{-t/\tau_s}$$



Difference of two exponentials

$$s(t) = e^{-t/\tau_{decay}} - e^{-t/\tau_{rise}}$$



Figures from Roth & van Rossum

http://homepages.inf.ed.ac.uk/mvanross/reprints/roth_mvr_chap.pdf

Summary

- We can mathematically and computationally model neurons and synapses at various levels of detail.
- The Hodgkin-Huxley model linked sodium and potassium currents to the action potential.
- Synaptic transmission is commonly modelling as a time varying conductance.